

Molar Mass Distribution and Chemical Composition Distribution of PS-*b*-PMMA Block Copolymers Determined by Diffusion Ordered Spectroscopy

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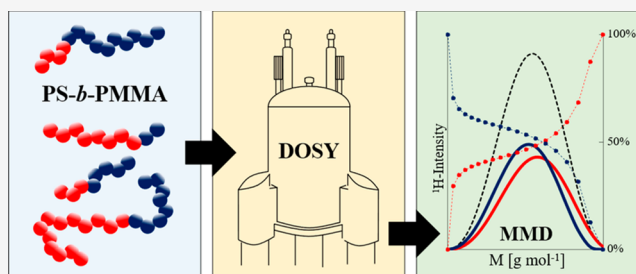


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Supporting Information

ABSTRACT: It is well-known that the molar mass distribution (MMD) of block copolymers will be performed by size exclusion chromatography (SEC) with two calibrated detectors to determine the molar mass distribution via the average chemical composition distribution (CCD) of the block copolymers. This work will present a new way to determine the molar mass distribution via the average chemical composition of block copolymers by using the nuclear magnetic resonance (NMR) based diffusion ordered spectroscopy (DOSY) with inverse Laplace transform (ILT). This DOSY approach allows the determination of the MMD of copolymers in the bulk sample. It will be shown that DOSY can be enabled for the determination of the MMD via the CCD of block copolymers by adopting the dual detector SEC approach to DOSY. Therefore, several PS-*b*-PMMA block copolymers were analyzed by this new DOSY approach as well as the dual SEC-UV-RI detection. The DOSY results of the MMD and CCD are in good agreement with the common SEC-RI-UV dual detector copolymer analysis.



1. INTRODUCTION

Block copolymers have a wide range of applications like drug delivery,¹ 3D printing,² and dispersion.³ The study of copolymers is very important because the chemical composition and chain length strongly influence the physical and chemical properties of these polymers.

Whereas ¹H NMR is typically used to determine the average chemical composition of block copolymers and to some extent the number-average molar mass M_n via the end group of the polymer, DOSY enables the separation of mixtures, allows for the determination of the hydrodynamic radius, and also for obtaining the weight-average molar mass M_w . The relation between the diffusion coefficient D and the weight-average molar mass M_w of the polymer is given by the scaling equation as follows.⁴

$$D = aM_w^b \quad (1)$$

The parameter b describes the solvent quality, with the Flory exponent $b \sim -0.6$ for a good solvent.

In the case of homopolymers even the determination of the molar mass distribution could be performed with DOSY including the calculation of the weight-average molar mass M_w and number-average molar mass M_n and PDI.^{5–16} Most works deal with M_w of polymers determined due to the relating calibration of the diffusion coefficient according to eq 1. Multiple methods like the gamma distribution model,^{7,12,14} inverse Laplace transform,^{11,13,16} CONTIN,^{15,17} DECRA,^{6,18} and many more were investigating molar mass distributions of

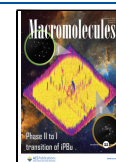
polymers. Viéville et al. successfully investigated the molar mass distribution of linear poly(ethylene oxide)s by using DOSY with inverse Laplace transform.¹³ Even PS-*b*-PEO block copolymers were studied by using diffusion NMR regarding their molar mass M_w ⁹ and its composition.¹⁹ Hiller separated PS-*b*-PMMA block copolymers from their homopolymer precursors with DOSY and analyzed the copolymers and homopolymers regarding M_w as well as their contents.²⁰ Junkers et al. investigated the solvent-independent M_w determination of polymers based on a truly universal calibration and also studied oligomer distributions applying diffusion ordered spectroscopy.^{21,22}

To provide a comprehensive copolymer characterization, further analysis has to be carried out. Copolymer analysis is commonly performed by using multidetector SEC with at least two hyphenated detectors, which are calibrated to quantitatively determine the chemical compositions during the elution. This is required for the calculation of the molar mass distribution with the relevant weight-average molar mass M_w , the number-average molar mass M_n , and the polydispersity index PDI.^{23,24} Striegel et al. have published several

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applications of multiple detection.^{25,26} Kilz and Pasch published a review.^{27,28} Further developments of multiple detection were also performed by Hiller et al. in comparison to online HPLC-NMR.^{23,24,29}

This paper will show that it is also possible to use DOSY for the determination of molar mass distributions of block copolymers by including the average chemical composition distribution. This approach is typically used in the SEC multidetector analysis. We will show that DOSY measurements can also provide M_w and M_n data for PS-*b*-PMMA block copolymers which will be comparable to the SEC dual detector analysis

2. EXPERIMENTAL SECTION

2.1. Samples. The molar masses and average chemical compositions of the PS-*b*-PMMA block copolymers are shown in Table 1. The molar masses of the homopolymer calibration standards

Table 1. Molar Masses and Average Chemical Composition of the PS-*b*-PMMA Block Copolymers Given by the Manufacturer

copolymer	M_w [g mol ⁻¹]	M_w (PS) [g mol ⁻¹]	M_w (PMMA) [g mol ⁻¹]	PS:PMMA
pp50k	50000	22000	28000	1:1.3
pp81k	81000	27000	54000	1:2
pp84k	84400	42200	42200	1:1
pp108k	108000	54000	54000	1:1

are summarized in Table S1 of the Supporting Information. All polymer samples were produced by PSS Polymer Standard Service GmbH (Mainz, Germany).

2.2. Size Exclusion Chromatography. An Agilent 1100/1200 HPLC system consisting of a quaternary pump, an autosampler, a column oven, and a refractive index (RI) and single wave UV ($\lambda = 254$ nm) detector equipped with a ResiPore column (Agilent Technologies) was used for the SEC-UV-RI analysis. All samples were dissolved in HPLC grade tetrahydrofuran (THF) from VWR Chemicals (Darmstadt, Germany) and were analyzed by injecting 100 μ L of a concentration of 1 mg mL⁻¹ and using a flow of 1 mL min⁻¹.

2.3. Diffusion Ordered Spectroscopy. Diffusion experiments were performed with a Bruker Avance III HDX 700 MHz spectrometer equipped with a 5 mm helium-cooled quadrupole resonance cryoprobe H(C,N,P). All diffusion experiments were performed with the Bruker dsteibpgp3s pulse sequence by using double stimulated echo for convection compensation and longitudinal eddy current delay (LED) with bipolar gradients for diffusion and three spoil gradient pulses.³⁰ The maximum gradient strength of the probe was calibrated to 56.9 G cm⁻¹ corresponding to the doped water diffusion coefficient of 1.91×10^{-9} m² s⁻¹ at 298.15 K.

All NMR samples were dissolved in 0.6 mL of deuterated dichloromethane (2 mg mL⁻¹). Deuterated CD₂Cl₂ was purchased from Deutero GmbH (Kastellaun, Germany).

The diffusion coefficients of the homopolymers were determined by varying the gradient strength in 32 linear steps between 10 and 80% of the maximum gradient strength.

The block copolymer diffusion coefficients were determined with a linear variation of 128 gradient strengths between 3 and 98% of the maximum gradient amplitude. The data are shown in Figures S1–S8.

The obtained signal attenuation was used to calculate the self-diffusion coefficients according to the Stejskal–Tanner equation:³¹

$$I = I_0 \exp\{-DQ\}$$
$$\text{with } Q = \gamma^2 g^2 \delta^2 \left(\Delta - \frac{\delta}{3} \right) \quad (2)$$

where D is the self-diffusion coefficient, I is the measured proton signal intensity for each gradient strength, I_0 is the unattenuated proton signal intensity, γ is the gyromagnetic ratio of the observed nucleus, g is the gradient strength, δ is the gradient duration, and Δ is the diffusion delay.

Copolymer diffusion data were acquired with a gradient pulse length (δ) of 2.8 ms and diffusion delays (Δ) between 140 and 220 ms and a 90° proton pulse of 7.3 μ s for all copolymers. All experiments were recorded by using 16 steady-state scans and 16 scans for each gradient strength. The acquisition time was 2.2 s with a relaxation delay of 3 s using 42616 data points for the F2 dimension.

2.4. Inverse Laplace Transform. Fourier transformation and phase and baseline correction were done by using TopSpin 4.1.4. Afterward, the inverse Laplace transform (ILT) was performed by using Bruker's Dynamics Center with the following parameters: logarithmic grid, second derivative regularization with 1×10^{-3} as regularization parameter, find alpha automatically, 128 grid points, and applied kernel compression.

3. RESULTS AND DISCUSSION

3.1. Homopolymer Calibration with DOSY. The standard procedure of SEC for the multidetector analysis of block copolymers is the molar mass calibration of the containing homopolymers. The same procedure should be applied now to the DOSY copolymer analysis. A 0.15% w/w homopolymer solution was prepared, and DOSY was measured for each PS and PMMA homopolymer of Table S1. The diffusion coefficients of these homopolymers in dependence of the molar mass M_w are shown in Figure 1.

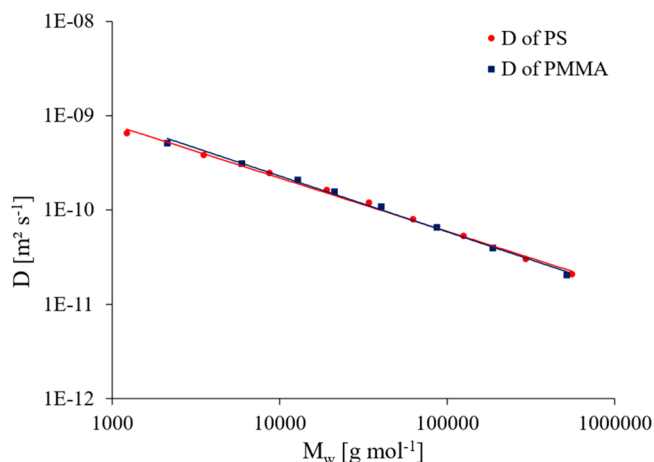


Figure 1. Double-logarithmic plot of average molecular weight M_w against diffusion coefficients D [m² s⁻¹] of PS (red) and PMMA (blue) determined by DOSY (solid lines are fitted molar mass dependences).

The logarithmic plot of the diffusion coefficient versus molar mass M_w shows linear dependences for the PS and PMMA homopolymers. The fitted parameters of factor a and exponent b of eq 1 are presented in Table 2.

Table 2. Fitted Parameters a and b According to Eq 1 for PS and PMMA Homopolymers

homopolymer	a	b	R^2
PS	4.026×10^{-8}	-0.566	0.996
PMMA	5.362×10^{-8}	-0.592	0.993

According to Figure 1 and Table 2, the data meet the solvent and concentration criteria for a good solvent indicated by the exponent b .³²

3.2. Copolymer Composition with DOSY. To determine the molar mass distributions of block copolymers with DOSY, we want to adopt the same approach of the SEC-UV-RI dual detector analysis to the DOSY studies. Therefore, four PS-*b*-PMMA copolymer solutions of different compositions and chain lengths were prepared for the copolymer analysis (Table 1). PS-*b*-PMMA copolymers were chosen because of the possibility to determine diffusion coefficients and signal intensities for each copolymer block separately by analyzing separated signals. The aromatic protons were chosen for the PS block, and the methoxy group was used for the PMMA block.

Self-diffusion coefficients of copolymers were determined by varying the gradient strength in 128 linear steps between 3 and 98% of the maximum gradient strength. The gradient strength and diffusion delays were chosen such as that the signal intensity of the methoxy group at 3.6 ppm was <1% for the strongest gradient (98%) compared to the signal intensity at 3% gradient strength. According to these preparation conditions, the 2D-DOSY experiments were performed. The results are illustrated in Figure 2 showing the PMMA methoxy group and the aromatic protons of PS.

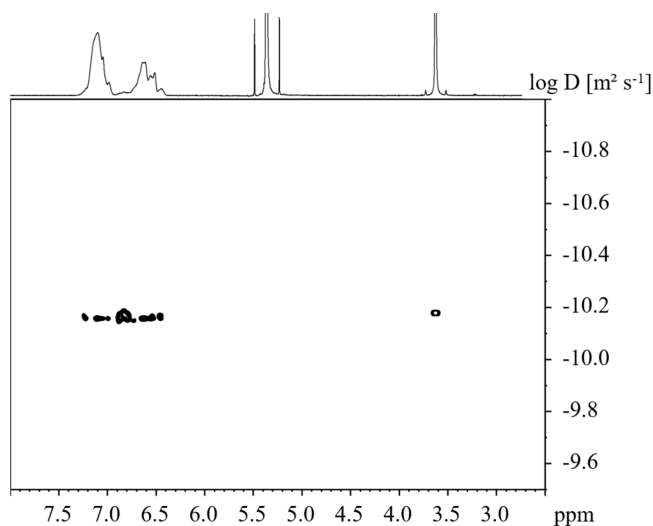


Figure 2. 2D-DOSY plot of PS-*b*-PMMA pp84k block copolymer showing the methoxy group (3.6 ppm) of PMMA and the aromatic protons (7.3–6.4 ppm) of PS.

The 2D-DOSY plot presents separated signals for each block. This is an ideal condition for the DOSY-ILT analysis as both signals of the aromatics and the methoxy group can be analyzed independently. After proton signal integrations of these two species at each gradient strength, the processing including the inverse Laplace transform was performed by using Bruker's Dynamics Center (Experimental Part 2.4). The obtained ILT intensities of the aromatics and methoxy group for each respective diffusion coefficient were then used to determine the weight-average chemical compositions at each diffusion coefficient.

The weight chemical compositions are the basic assumption for the calculation of the molar mass distribution of block copolymers as seen below. In the case of the SEC-UV-RI dual detector analysis, the weight chemical compositions are

calculated by using so-called response factors of the two detectors. The measured response factors of the UV and RI detectors are shown in Table S5. The equations and experimental data for determining these factors are given in Table S7 and Figure S10. The UV and RI chromatograms of the copolymers are shown in Figure S11.

In the case of DOSY, however, the weight fractions of the PS and PMMA blocks can be directly derived from the integrated molar fractions of the DOSY-ILT intensities according to eq 3:

$$\text{wt}_{\text{PS}}(D_i) = \frac{\frac{I_{\text{raw-PS}}(D_i)}{n_{\text{PS}}} M_{\text{PS}}}{\frac{I_{\text{raw-PS}}(D_i)}{n_{\text{PS}}} M_{\text{PS}} + \frac{I_{\text{raw-PMMA}}(D_i)}{n_{\text{PMMA}}} M_{\text{PMMA}}}$$

with $\text{wt}_{\text{PMMA}}(D_i) = 1 - \text{wt}_{\text{PS}}(D_i)$ (3)

where I_{raw} are the obtained DOSY-ILT signal intensities for the PS and PMMA blocks from the raw data in Figure S11, n is the number of protons of the appropriate signal for one monomer unit, and M is the molar mass of the corresponding homopolymer monomer unit. Further equations for DOSY are given in Table S6.

The results of all investigated PS-*b*-PMMA by the DOSY-ILT block copolymer analysis compared to the SEC-UV-RI dual detector analysis are illustrated in Figure 3.

The comparison shows the same behavior for the chemical compositions for both methods even considering the lower DOSY-ILT resolution. The diffusion experiments were acquired at 128 gradient strengths. The ILT was performed for each PS-*b*-PMMA copolymer and yields about 20 data points per copolymer compared to ~100 data points by using the SEC-UV-RI dual detector analysis. The copolymers pp50k, pp81k, and pp84k polymers also contain remaining PS homopolymer. This is nicely indicated for both the DOSY-ILT and the SEC-UV-RI analysis. Looking at the diffusion data, the PS intensity curve (red) is broader than the PMMA intensity curve (blue) and is also ending at larger diffusion coefficients than the PMMA curve. This can only be caused by a second component within the lower molar mass range of M_w besides the main copolymer molar masses. A pure block copolymer would lead to just one diffusion coefficient for both blocks, as these belong to the same macromolecule. These remaining homopolymer components lead to larger diffusion coefficients, but the data processing cannot separate both components because the differences of these molar masses are too small to completely determine the diffusion coefficient for each component.^{33,34} The same could be observed for the SEC-UV-RI analysis where a little homopolymer component can be observed at larger retention times. Figures S12 and S13 also show the simulated UV and RI chromatograms for the individual PS and PMMA blocks. These chromatograms are derived from the difference of the total UV or RI chromatograms and the individual UV or RI parts of PS and PMMA which are given by the product of the corresponding response factor and the chemical compositions, respectively. These figures also demonstrate for the copolymers pp50k, pp81k, and pp84k that the RI and UV signal intensities are still detected for PS when the PMMA signal intensity almost dropped to zero. The aromatic system of polystyrene shows UV absorption at 254 nm. At the same wavelength, PMMA shows almost no absorption which is supported by the response factors as high values indicate strong absorption (Table S5). Therefore, the UV response in Figure S13 is

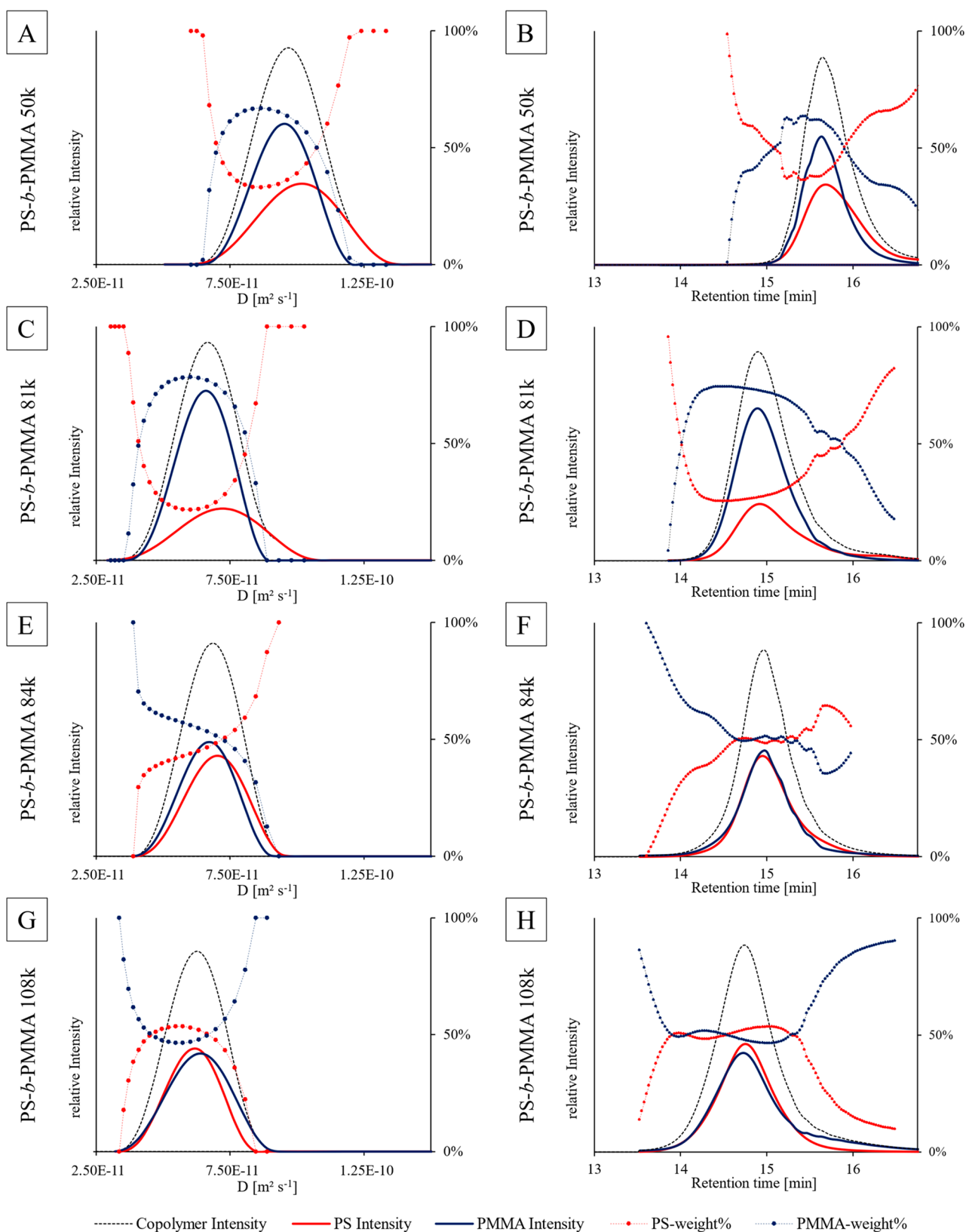


Figure 3. Chemical composition distributions of the PS-*b*-PMMA copolymers analyzed by DOSY-ILT (left: A, C, E, G) and SEC-UV-RI dual detector analysis (right: B, D, F, H) for the determination of the molar mass distributions.

plotted at about 100 times higher intensity as in Figure S12 to show clearly the UV signal of PMMA. A significant difference in response factors is the precondition for the SEC-UV-RI copolymer analysis due to a reliable quantification of the

chemical compositions of both copolymer blocks during the elution. The data of the largest PS-*b*-PMMA (pp108k) copolymer indicate opposite results regarding the remaining homopolymer as both analytical methods show that the PS

Table 3. Molar Mass Distributions of the Investigated PS-*b*-PMMA Copolymers Analyzed by SEC-UV-RI and DOSY-ILT^a

PS- <i>b</i> -PMMA	method	copolymer			PS signals			PMMA signals		
		M_n	M_w	PDI	M_n^T	M_w^T	PDI	M_n^T	M_w^T	PDI
pp50k	SEC-UV-RI	45500	48600	1.07						
	DOSY-ILT	42200	44500	1.05	39900	42400	1.06	45200	46800	1.03
pp81k	SEC-UV-RI	82500	88200	1.07						
	DOSY-ILT	81100	89400	1.10	75200	86300	1.15	85900	91700	1.07
pp84k	SEC-UV-RI	78600	84300	1.07						
	DOSY-ILT	79200	84800	1.07	77200	83000	1.07	82700	87900	1.06
pp108k	SEC-UV-RI	85900	97900	1.14						
	DOSY-ILT	94400	102000	1.08	97000	105000	1.08	90900	100000	1.10

^aDOSY signals of the PS and PMMA blocks were analyzed independently to determine the true molar mass distribution of the copolymers.

intensity drops to zero before the PMMA intensity does, so there is a PMMA homopolymer contamination (see Figure 3G,H and Figures S12D,H and S13D,H).

3.3. Molar Mass Distribution with DOSY. After the successful determination of the chemical compositions with DOSY-ILT the calculation of the molar mass distributions can be performed by using the same approach as the SEC-UV-RI analysis. In this case, an individual calculation of the copolymer molar masses will be executed by including the chemical composition dependences as well as the molar mass calibrations of both homopolymers. Therefore, the following equations were used to calculate the molar mass distributions of the copolymers. The approach for calculating the $M_{\text{Copolymer}}$ values of the copolymers for DOSY and the SEC-UV-RI analysis is described by eq 4:

$$\log M_{\text{Copolymer}}(x) = \text{wt}_{\text{PS}}(x) \log M_{\text{PS}}(x) + \text{wt}_{\text{PMMA}}(x) \log M_{\text{PMMA}}(x) \quad (4)$$

with $x = D_i$ or Rt_i

where the wt %_{PS} and wt %_{PMMA} are taken from Figure 3 at D_i as the diffusion coefficients obtained by DOSY-ILT or Rt_i as the retention time from the SEC-UV-RI analysis as well as the molar masses M_{PS} and M_{PMMA} of both homopolymers PS and PMMA calculated by eq 5 (adopted from eq 1) at a certain D_i for DOSY and the polynomial eq 6 at a certain Rt_i for the SEC:

$$M_{\text{Homopolymer}}(D_i) = \left(\frac{D_i}{a}\right)^{1/b} \quad (5)$$

$$\log M_{\text{Homopolymer}}(Rt_i) = c_1 Rt_i^5 + c_2 Rt_i^4 + c_3 Rt_i^3 + c_4 Rt_i^2 + c_5 Rt_i + c_6 \quad (6)$$

Equations 5 and 6 must be performed for PS and PMMA separately. The parameters for eq 5 are given in Table 2, and the parameters of eq 6 are derived from Figure S9 and given in Table S4.

Finally, the number- and weight-average molar masses can be calculated with the well-known eqs 7 and 8:

$$M_n = \frac{\sum_i I_{\text{Copolymer}}(x)}{\sum_i \left(\frac{I_{\text{Copolymer}}(x)}{M_{\text{Copolymer}}(x)} \right)} \quad (7)$$

$$M_w = \frac{\sum_i [I_{\text{Copolymer}}(x) M_{\text{Copolymer}}(x)]}{\sum_i I_{\text{Copolymer}}(x)} \quad (8)$$

$$I_{\text{Copolymer}}(x) = I_{\text{PS}}(x) + I_{\text{PMMA}}(x) \quad (9)$$

with $x = D_i$ or Rt_i

where $I_{\text{Copolymer}}$ is the intensity of the copolymer, which is either the sum of the obtained normalized DOSY-ILT intensities of the PS and PMMA blocks at the given diffusion coefficient D_i or the sum of the normalized SEC intensities of both blocks at the given retention time Rt_i , respectively. The normalized intensities I_{PS} and I_{PMMA} for the PS and PMMA blocks are shown as red solid lines for PS and blue solid lines for PMMA in Figure 3. The copolymer intensities are demonstrated as the black dashed lines in Figure 3. The corresponding equations for the calculation of the molar mass distributions with DOSY and SEC are also summarized in Tables S6 and S7.

DOSY-ILT has one great advantage over the dual detector analysis as each block is represented by signals with a unique chemical shift, which can be analyzed independently. SEC dual detector analysis allows for investigating both blocks, but these detectors usually detect both homopolymers with different responses simultaneously. In most cases, there is no possibility to determine the blocks completely independent from each other with SEC.

Both analytical approaches revealed homopolymer contaminations for each PS-*b*-PMMA copolymer, but DOSY-ILT can be used to determine a true or at least more accurate molar mass distribution. Therefore, the ¹H signal of the block that shows no homopolymer contamination is used to calculate M_n^T and M_w^T with the corresponding calibration that represents the larger block of the copolymer. As the diffusion coefficient distributions do not contain smaller homopolymer, which would increase the average diffusion coefficient, the new calculated molar mass distributions should indicate greater M_n and M_w values. The molar mass distribution values of the block which also presents the homopolymer contamination are illustrated in italic as these values do not represent the copolymer correctly.

All results are listed in Table 3 and compared to the SEC-UV-RI copolymer analysis data.

When SEC-UV-RI data are compared with the corresponding DOSY-ILT results, simply stated, they match. Some polymers show a better consistency than others, but the chemical compositions as discussed before as well as the molar mass distributions are agreeing very well. Considering for instance PS-*b*-PMMA pp84k (42000/42000), it shows almost the same number- and weight-average molar masses as the SEC-UV-RI analysis. This polymer has the smallest contamination of PS homopolymer, which may be the reason for the

almost identical molar mass distribution and matches the given average molar mass data of the manufacturer. The pp81k instead contains more PS homopolymer, which may lead to deviations of the manufacturer's M_w distribution but shows the same results for the chromatographic and DOSY-ILT analysis. The pp50k and pp108k have the largest deviations from the stated weight distributions but still show similar results regarding DOSY-ILT and SEC-UV-RI block copolymer analysis. In addition, the chemical compositions can now also be shown in dependence of the molar masses. These data are shown in Figure S14 and demonstrate the same behavior for both analytical methods. Only small shifts of the chemical compositions versus molar mass are visible for the different copolymers. In general, it can be noted that the chemical compositions determined with DOSY-ILT reveal the same trend as the SEC-UV-RI analysis.

Furthermore, comparing the M_p values of both analytical approaches, the M_p data of the component without homopolymer contamination should be chosen over the other block as the remaining homopolymer shifts the diffusion coefficient to higher diffusion values.

The question remains which approach is the more accurate one. The traditional SEC has proven that it is a reliable and well-established analytical method.^{35–39} The DOSY-ILT approach instead shows accurate results compared to SEC dual detector analysis even if the inverse Laplace transform is just a mathematical interpretation of the acquired data. There is no need of choosing the right column, temperature, flow, and appropriate detector as the diffusion experiments are prepared fast and easy after the weighed sample is solved and filled into the NMR tube. The experiment time for a DOSY may be longer than the time for chromatographic separation, but even with 128 increments used, it was no longer than 3 h.

4. CONCLUSION

The DOSY-ILT block copolymer analysis approach shows excellent results as verified by the traditional SEC dual detector analysis and allows a fast and easy copolymer characterization. No second detector is needed as proton diffusion NMR experiments deliver separated chemical shifts for the PS and the PMMA block. The determined chemical compositions as well as molar mass distributions are comparable to the results of the traditional chromatographic approach, even if the resolution is lower. The main advantage of using DOSY-ILT over SEC-dual detector analysis is the determination of the true molar mass distribution as the block, which does not contain homopolymer contamination, reflects the true diffusion coefficient of the block copolymer.

■ ASSOCIATED CONTENT

SI Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acs.macromol.2c01505>.

Additional experimental details, materials, tables, figures, and equations (PDF)

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Notes

The authors declare no competing financial interest.

■ DEDICATION

Dedicated to Professor Harald Pasch on the occasion of his 70th birthday.

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